

# Abdulrahman Diaa

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## Awards

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|-----------------------------------------------------------------------------|---------------------------------------------------------|
| 2026-2027: NSERC Canada Graduate Research Scholarship - Doctoral - CA\$120K | University of Waterloo                                  |
| 2026-2027: Interop Labs PhD Fellowship - CA\$110K                           | University of Waterloo                                  |
| 2026-2027: President's Graduate Scholarship - CA\$10K                       | University of Waterloo                                  |
| 2024 : Best Poster Award - CA\$1K                                           | Cybersecurity and Privacy Institute Conference (CPI'24) |
| 2021-2025: David R. Cheriton Graduate Scholarship - CA\$40K                 | University of Waterloo                                  |
| 2021-2023: International Masters Award of Excellence - CA\$12.5K            | University of Waterloo                                  |
| 2021-2022: Johnston International Entrance Scholarship - CA\$5K             | University of Waterloo                                  |
| 2016-2021: Tarek Nour AUC Scholarship - US\$100K                            | The American University in Cairo                        |

## Research Experience

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| <b>Student Researcher</b><br><i>Google</i><br><b>Topics:</b> Private RAG and Nearest Neighbor Search<br><b>Reference:</b> <a href="#">Sarvar Patel</a>                                                                                                                                                  | <b>Ontario, Canada</b><br>Apr 2026 - Dec 2026         |
| <b>Cybersecurity Researcher</b><br><i>Airbus Defence and Space – VCX</i><br><b>Topics:</b> Practical Batch Private Information Retrieval and Privacy-preserving DBSCAN<br><b>Reference:</b> <a href="#">Dr. Erik-Oliver Blass</a>                                                                       | <b>München, Germany</b><br>Jan 2025 - July 2025       |
| <b>Research Intern</b><br><i>École Polytechnique Fédérale de Lausanne – Lab for Data Security (LDS)</i><br><b>Topic:</b> Privacy-preserving Federated Learning for Recurrent Neural Networks<br><b>Reference:</b> <a href="#">Prof. Jean-Pierre Hubaux</a>                                              | <b>Lausanne, Switzerland</b><br>July 2021 - Sept 2021 |
| <b>Undergraduate Data Science Research Intern</b><br><i>Dell Technologies – Data Office Team</i><br><b>Topics:</b> Technology Forecasting and Distributed Neural Networks<br><b>Reference:</b> <a href="#">Eng. Steve Todd</a>                                                                          | <b>Cairo, Egypt</b><br>Feb 2020 - June 2021           |
| <b>Undergraduate Researcher</b><br><i>The American University in Cairo</i><br><b>Topic:</b> Deep Reinforcement Learning for Traffic Control<br><b>Reference:</b> <a href="#">Dr. Cherif Salama</a>                                                                                                      | <b>Cairo, Egypt</b><br>Sept 2019 - Jan 2020           |
| <b>Research Software Development Engineering Intern</b><br><i>Microsoft Advanced Technology Lab in Cairo – Document Knowledge Extraction Team</i><br><b>Topic:</b> Hierarchical Clustering for Schema-Inference from Semi-Structured Documents<br><b>Reference:</b> <a href="#">Eng. Achraf Chalabi</a> | <b>Cairo, Egypt</b><br>Jul 2019 - Sept 2019           |

## Publications

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- [1] **A. Diaa**, T. Aremu, and N. Lukas. “Optimizing Adaptive Attacks against Watermarks for Language Models”. In: The 42nd International Conference on Machine Learning. 2025.  
🏆 Spotlight paper at ICML'25 (Top 2.6%)  
🏆 Oral presentation at The 1st Workshop on GenAI Watermarking at ICLR'25 (Top 3.9%)  
🏆 Best poster award at Cybersecurity and Privacy Institute Conference CPI'24.
- [2] **A. Diaa**, T. Humphries, and F. Kerschbaum. “FastLloyd: Federated, Accurate, Secure, and Tunable  $k$ -Means Clustering with Differential Privacy”. In: The 34th USENIX Security Symposium. 2025.
- [3] **A. Diaa**, J. Petit, and F. Kerschbaum. *Semantic Watermarking with Order-Robust Detection over Sub-sentence Units*. arXiv preprint arXiv:2608.27666. 2026.

- [4] **A. Diaa** et al. "Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions". In: The 33rd USENIX Security Symposium. 2024.
- [5] N. Lukas, **A. Diaa**, L. Fenaux, and F. Kerschbaum. "Leveraging Optimization for Adaptive Attacks on Image Watermarks". In: The 12th International Conference on Learning Representations. 2024.
- [6] R. Mahdavi, **A. Diaa**, and F. Kerschbaum. "ZipPIR: High-throughput Single-server PIR without Client-side Storage". In: The 35th USENIX Security Symposium. 2026.
- [7] S. Sav, **A. Diaa**, A. Pyrgelis, JP. Bossuat, and JP. Hubaux. "Privacy-Preserving Federated Recurrent Neural Networks". In: The 23rd Privacy Enhancing Technologies Symposium. 2023.

## Patents

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**Dell Technologies – Patent No: US 11934487B2 (Granted)**

*Splitting Neural Networks on Multiple Edge Devices to Train on Vertically Distributed Data* Mar 2024

**Dell Technologies – Application No: US 17/160782 (Pending)**

*Forecasting Technology Phases Using Unsupervised Clustering with Wardley Maps* Jan 2021

## Education

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**University of Waterloo – CrySP Lab**

**Ontario, Canada**  
Sept 2021 - Aug 2027

PhD in Computer Science (Sept 2023 - Aug 2027) – **GPA: 95.5%**

**Thesis:** *On the Privacy and Traceability of Machine Learning Systems*

MMath in Computer Science (Sept 2021 - Aug 2023) – **GPA: 96.75%**

**Thesis:** *Differentially-private multiparty Clustering*

**Supervisor:** Prof. Florian Kerschbaum

**The American University in Cairo**

**Cairo, Egypt**  
Sept 2016 - June 2021

Bsc in Mathematics (Sept 2017 - June 2021) – **GPA: 4.0/4.0**

**Thesis:** *Lattice-based Cryptography and Fully Homomorphic Encryption: A Survey*

Bsc in Computer Engineering (Sept 2016 - June 2021) – **GPA: 4.0/4.0**

**Thesis:** *Mixture of Experts for Human Activity Classification from Images*

**Supervisors:** Dr. Ahmed El-Guindy, Dr. Cherif Salama and Dr. Mohammed Moustafa

## Academic Service

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**2026:** Reviewer *40th Annual Conference on Neural Information Processing Systems (NeurIPS)*

**2025:** Workshop Reviewer *1st Workshop on GenAI Watermarking (WMark@ICLR)*

**2025:** Artifact Reviewer (Ninja Award) *34th USENIX Security Symposium*

**2024:** Journal Reviewer *Transactions on Knowledge and Data Engineering (TKDE)*

## Extra-Curricular Activities

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**2017:** Academic Vice-president *Robotics Club, The American University in Cairo*

**2017:** ROV Control Engineer *Robotics Club, The American University in Cairo*

**2017:** Robotics Instructor *Robotics Club, The American University in Cairo*

**2017:** Content Editor *Geniuses (TV Show)*

**2016:** Helped raise ~4M USD for public schools *Geniuses (TV Show)*

**2016:** Geniuses (TV Show) Team Leader *Al-Bashaer International Schools*

## Teaching Experience

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### Teacher Assistant

*University of Waterloo*

**Course:** Privacy, Cryptography, Network, and Data Security

**Reference:** Florian Kerschbaum

**Ontario, Canada**

*Jan 2025 - Apr 2025*

### Teacher Assistant

*University of Waterloo*

**Course:** Privacy, Cryptography, Network, and Data Security

**Reference:** Thomas Humphries

**Ontario, Canada**

*Jan 2023 - Apr 2023*

### Teacher Assistant

*The American University in Cairo*

**Course:** Embedded Systems Laboratory

**Reference:** Dr. Suzanne Safwat

**Cairo, Egypt**

*Jan 2021 - June 2021*

### Teacher Assistant

*The American University in Cairo*

**Course:** Linear Algebra

**Reference:** Dr. Ahmed El-Guindy

**Cairo, Egypt**

*Sept 2020 - June 2021*

### Teacher Assistant

*The American University in Cairo*

**Course:** Programming Fundamentals

**Reference:** Dr. Howaida Ismail

**Cairo, Egypt**

*Sept 2017 - Jun 2019*

## Other Roles

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**2018-2019:** Student Technology Assistant

*C. of Learning & Teaching, The American University in Cairo*

**2015-2017:** A-Levels Teacher Assistant

*Al-Bashaer International Schools*

## Other Honors

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**2021:** President Cup

*The American University in Cairo*

**2016-2021:** Dean's list of Academic Standing

*The American University in Cairo*

**2020:** Challenge Coin for Innovation

*Dell Technologies*

**2019-2020:** Finalist

*Egyptian Collegiate Programming Contest*

**2016:** Best Player Cup

*Geniuses (TV-Show), Season 1*

## Engineering Experience

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### Software Engineering Intern

*Nokia EG*

**Cairo, Egypt**

*Aug 2018 - Sept 2018*

### Software Engineering Intern

*Microsoft EG*

**Cairo, Egypt**

*Jul 2017 - Aug 2017*

## Skills

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**Languages:** Python, C++, GoLang, C#, C, Rust, R, Verilog, SQL

**Frameworks:** Lattigo, CUDA, PyTorch, PySyft, OpenCV, ROS, SUMO, Arduino